MT1011 Engineering Materials

Crystal Defects

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Defects in crystals

- All crystalline material have defects.
- **Point defects** include impurity atoms/ions.
- A missing atom, **vacancy**, is another point defect.
- **Line defects** or **dislocations** are lines of defects.
- **Planar defects** are the boundaries between crystalline regions.
- Some materials are non-crystallineor amorpous (e.g. glass).

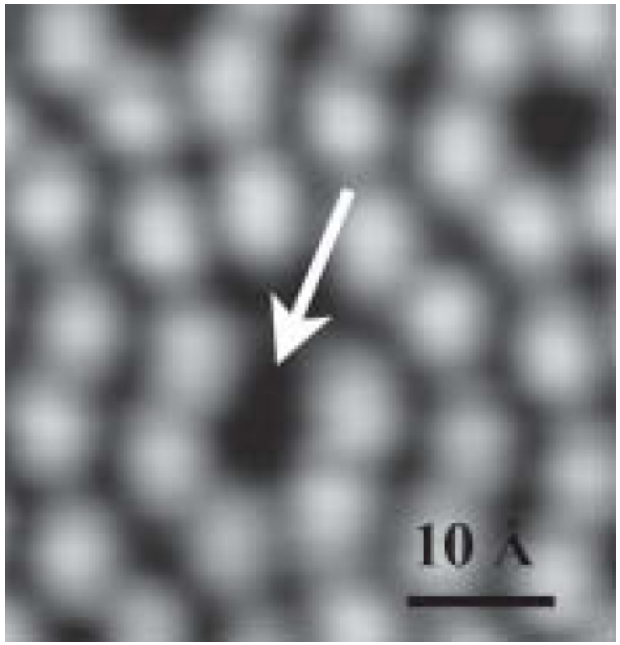
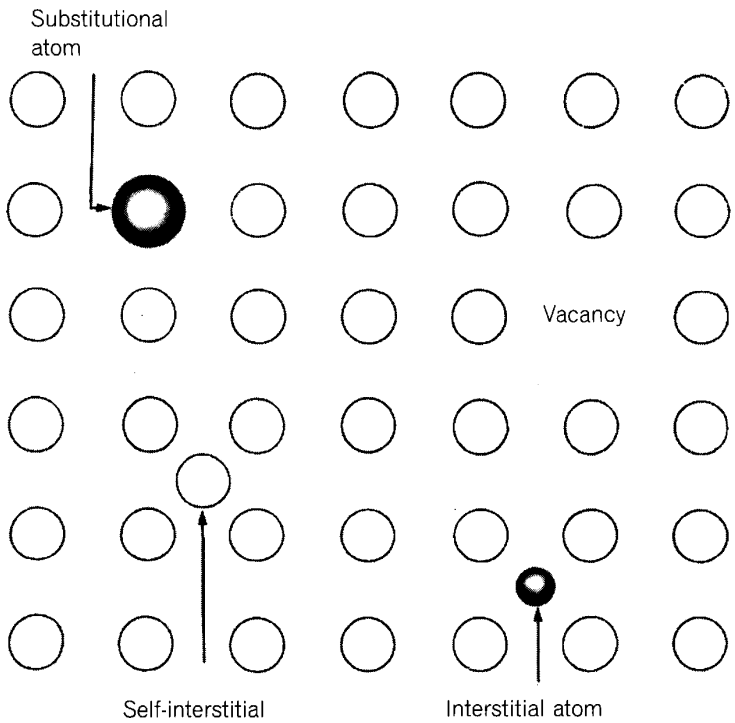


Figure: A vacancy on a (111)-type surface plane for silicon.

D. Huang, Stanford University, Calister & Rethswisch 2014

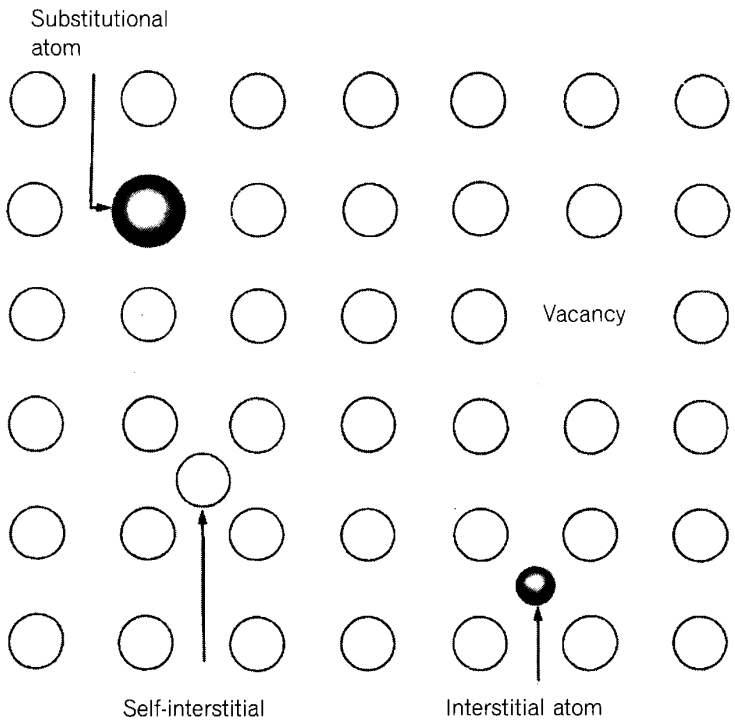
Point defects

- Point defects have **atomic** dimensions in all crystallographic directions.
- A foreign atom (the **solute**), can be an impurity atom or a deliberate alloying addition
- They can stay in one of two distinct positions in the parent crystal (the **solvent**).



Point defects

- If the solute **substitutes** for an atom of the solvent it is said to be a **substitutional** atom
- If the solute occupies a **hole** or **interstice** in the parent lattice it is called an **interstitial** atom.
- Which type a solute atom takes depends on its size relative to the solvent atoms.



Point defects

- Small solute atoms exist as an interstitials.
- In metal,s the most important interstitials are C, N and O.
- These are small atoms of less than 0.08 nm radius.

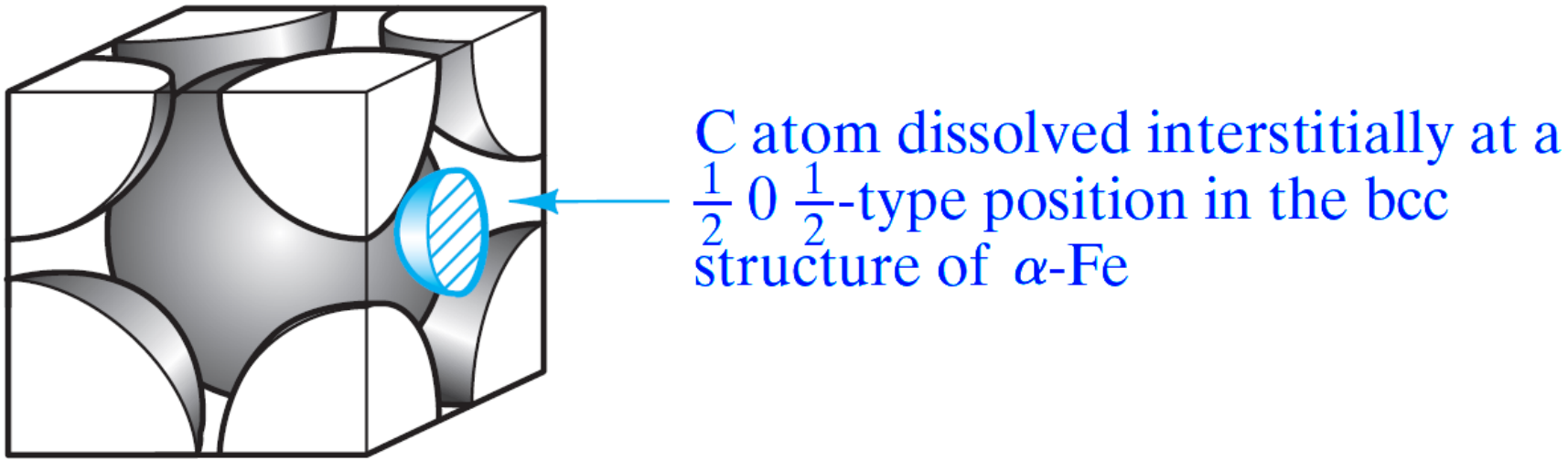


Figure: C interstitial in Fe lattice

- Vacancy is the simplest point defect
- It is a vacant lattice site or a missing atom.
- All crystalline solids have vacancies.
- Equilibrium number of vacancies N_v increase with temperature according to:

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

Q_v Energy required to form a vacancy

N Total number of atomic sites

T Absolute temperature in K

k The gas or Boltzmann's constant 1.38×10^{-23} J/K per atom, or 8.62×10^{-5} eV/K per atom



$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

- N_v increases exponentially with T
- For most metals, the fraction of vacancies Nv/N just below the melting temperature is around 10^{-4}
- One lattice site out of $10\,000$ will be empty.

Problem

Calculate the equilibrium number of vacancies per cubic meter for copper at $1000\,^{\circ}\mathrm{C}$. The energy for vacancy formation is $0.9\,\mathrm{eV}$ per atom. The atomic weight and density at $1000\,^{\circ}\mathrm{C}$ for Cu are $63.5\,\mathrm{g/mol}$ and $8.4\,\mathrm{g/cm^3}$, respectively.



- Unoccupied small spaces: **interstitial sites**.
- **Self-interstitial**: an atom from the crystal moved into an interstitial site.
- In metals, these give *large* distortions in the lattice (crowding).

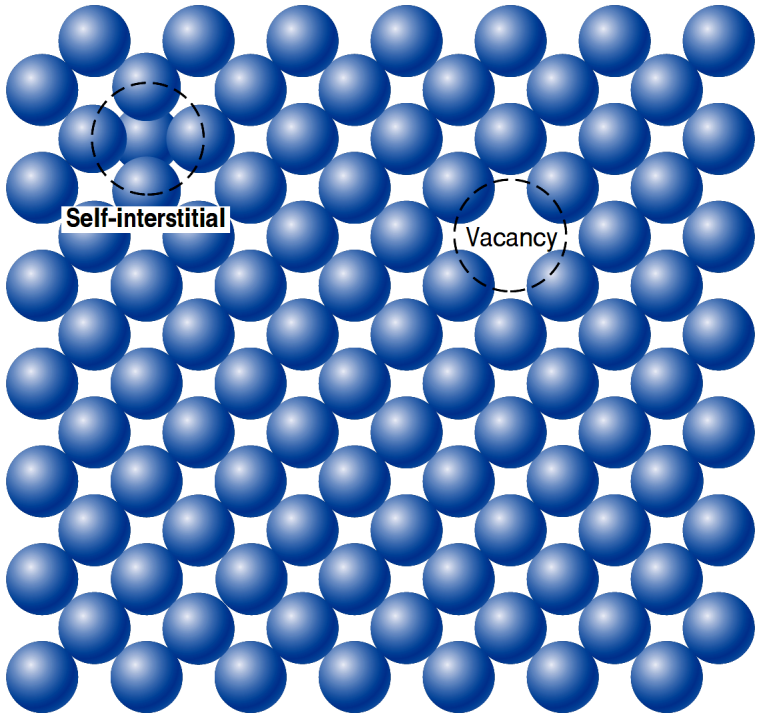


Figure: Self-interstitials and vacancy point defects

- The defect is **not** very probable in metals.
- Their concentrations are much lower compared to vacancies.
- They are more common in more open structures like Si.

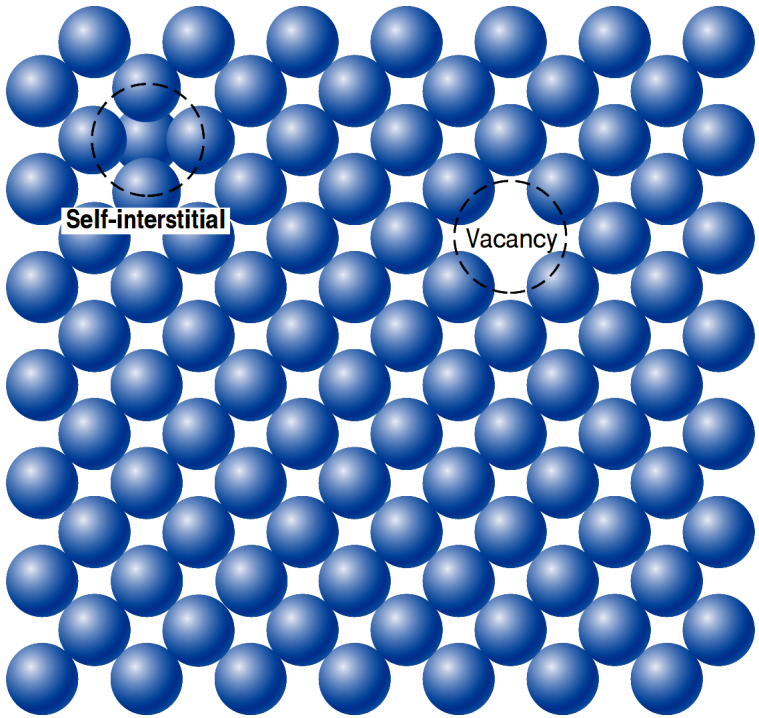


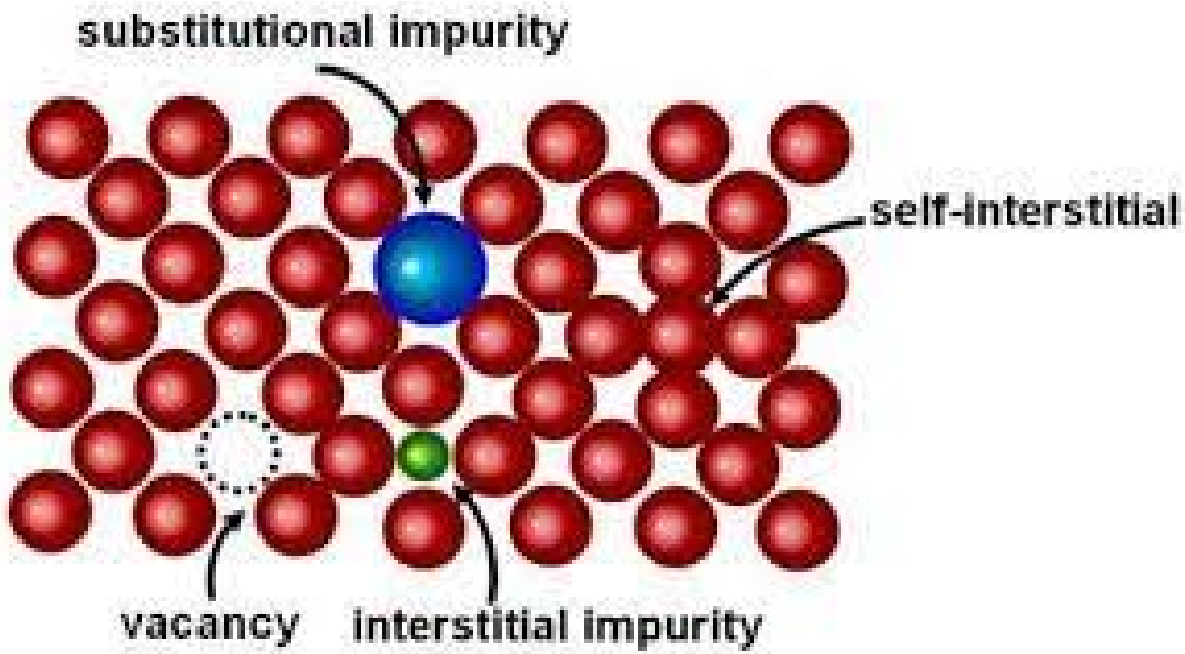
Figure: Self-interstitials and vacancy point defects

Point defects

Self interstitials

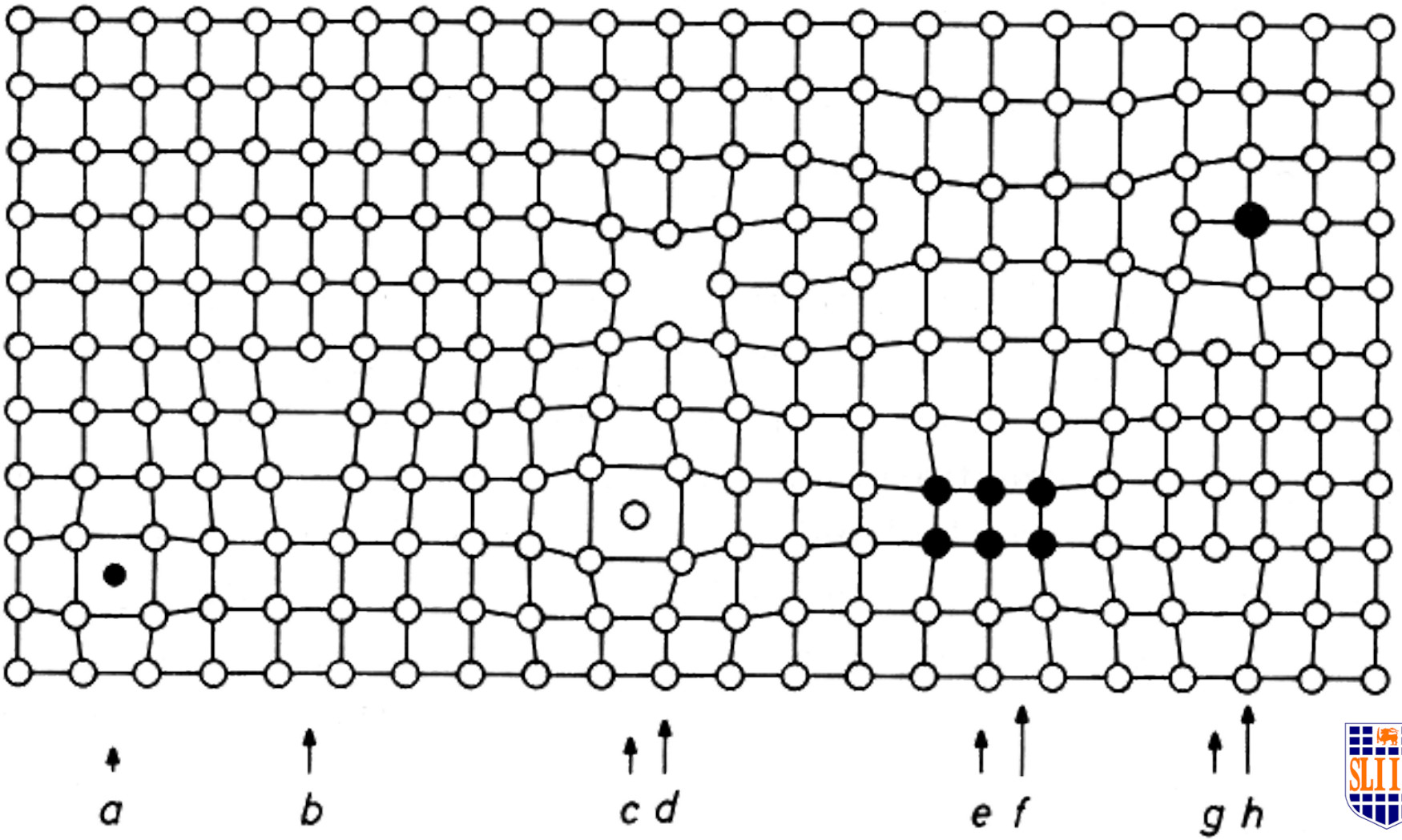
Types point defects in metals are

- Substitutional impurities
- Interstitial impurities
- Vacancies
- Self interstitials



Defects

Various types of defects



Line defects

Dislocations

- Line defects are long in one direction
- They measure only a few atomic diameters in width.
- They are mainly the dislocations.

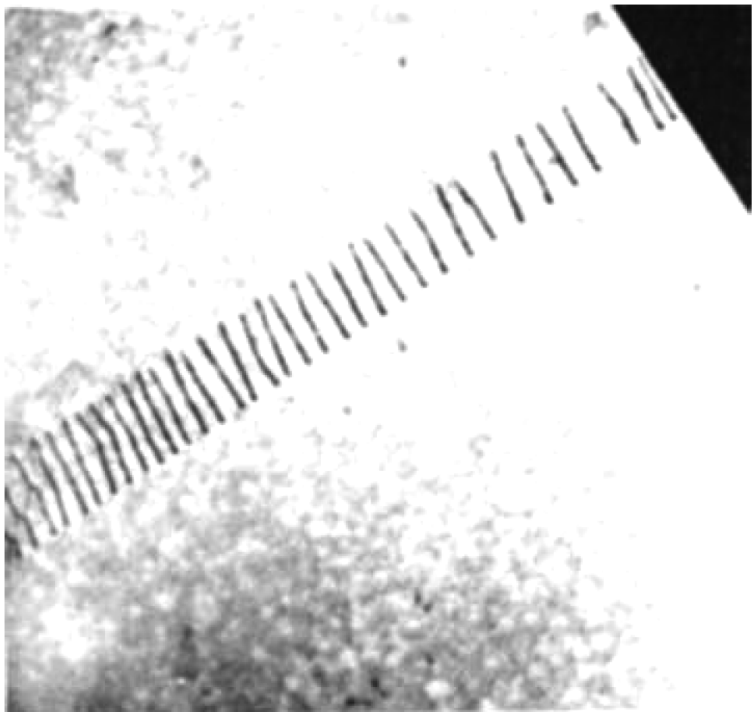


Figure: Dislocation arrays in an Al crystal

- Dislocations can explain the large difference between the **theoretical stress** (very high) needed to deform a perfect crystal plastically and the **actual stresses** measured in imperfect crystals, which are much lower.
- Dislocations were first predicted to explain the difference.

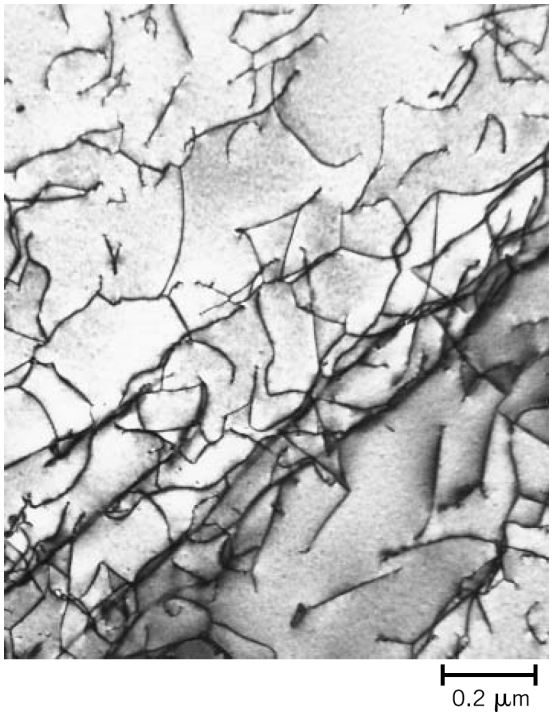


Figure: Dislocations (dark lines) in a Ti alloy.

- **Edge** and **screw** are the two main dislocation types.
- In an edge dislocation, lattice is distorted along the end of an extra half-plane of atoms (dislocation line)

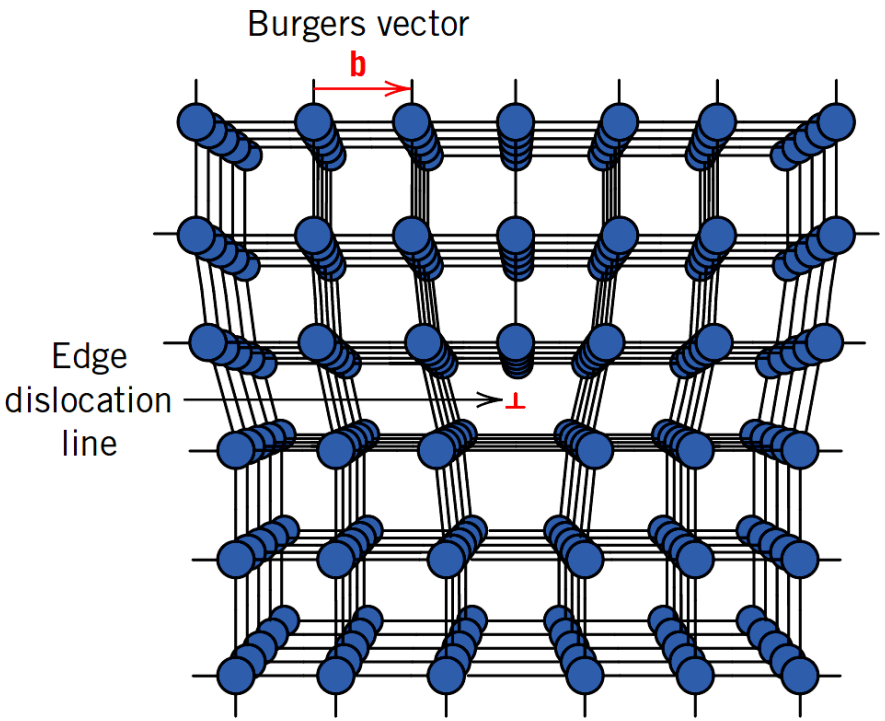


Figure: Edge dislocation

- A screw dislocation is similar to a shear distortion.
- Their dislocation line passes through the center of a spiral.

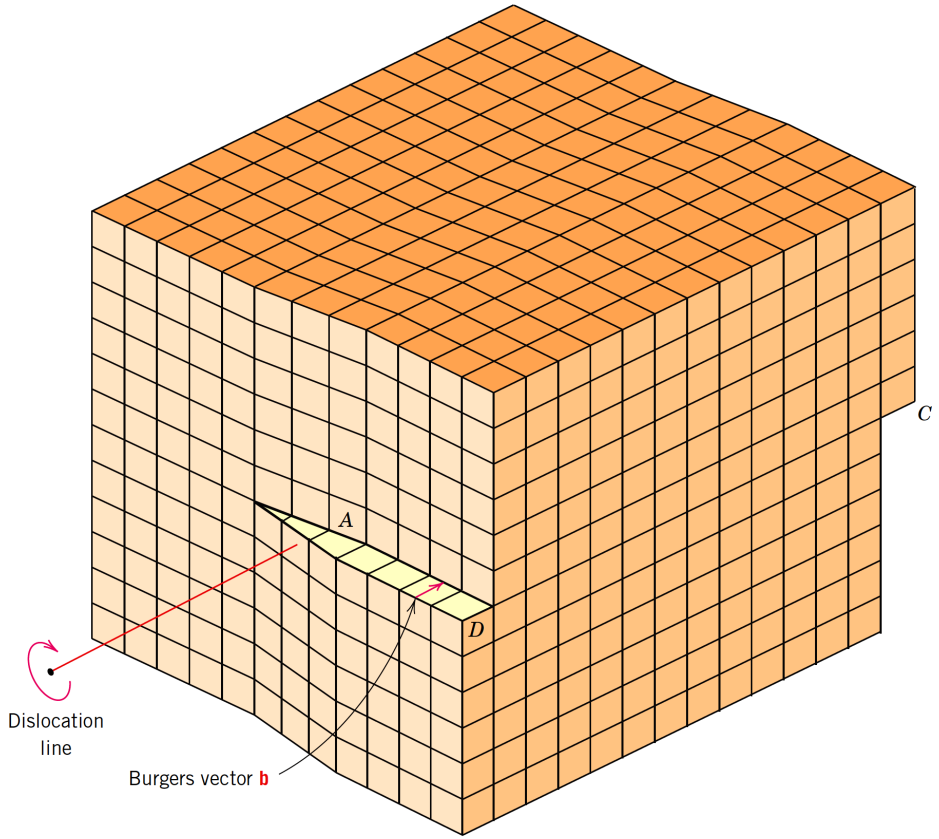
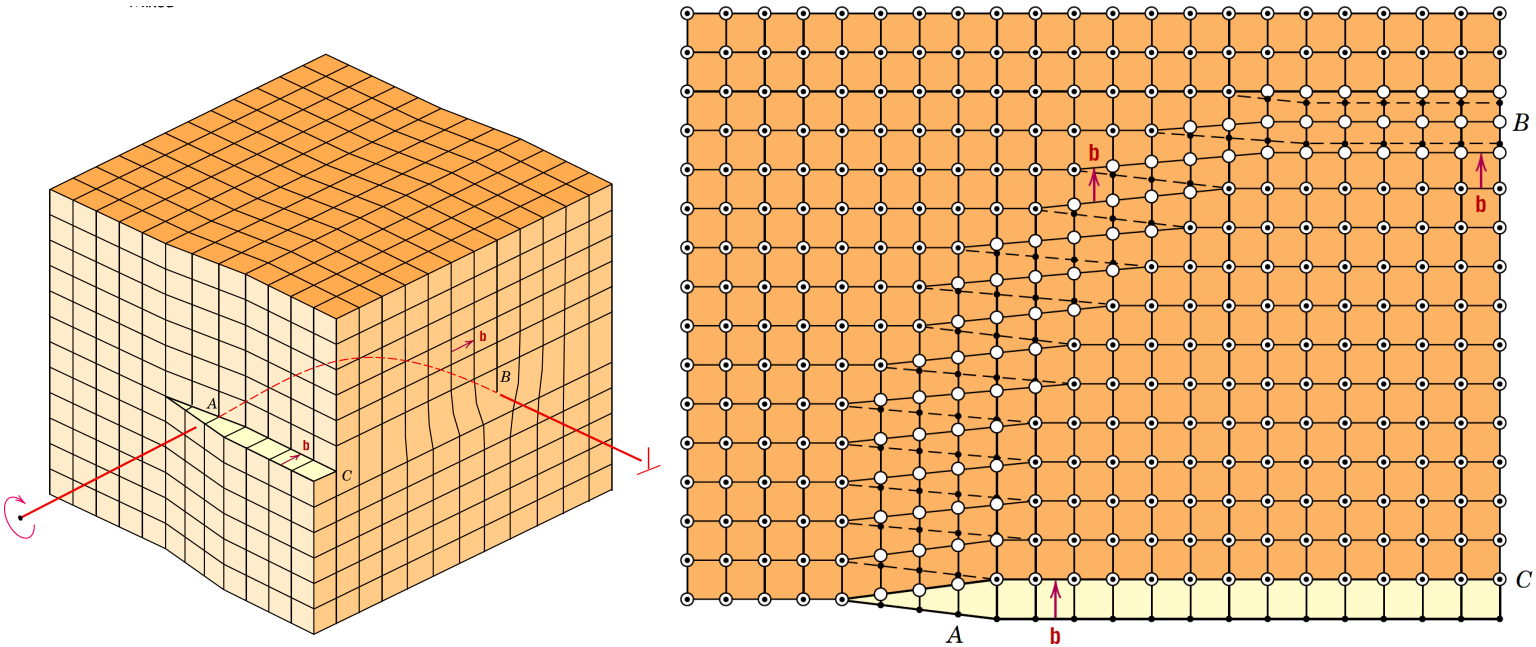


Figure: Screw dislocation

Many dislocations have both edge and screw parts (mixed dislocations).



Line defects

Dislocation movement

- Plastic deformation is due to the movement of large numbers of dislocations.
- An edge dislocation moves due to a shear stress applied in a direction perpendicular to its line.

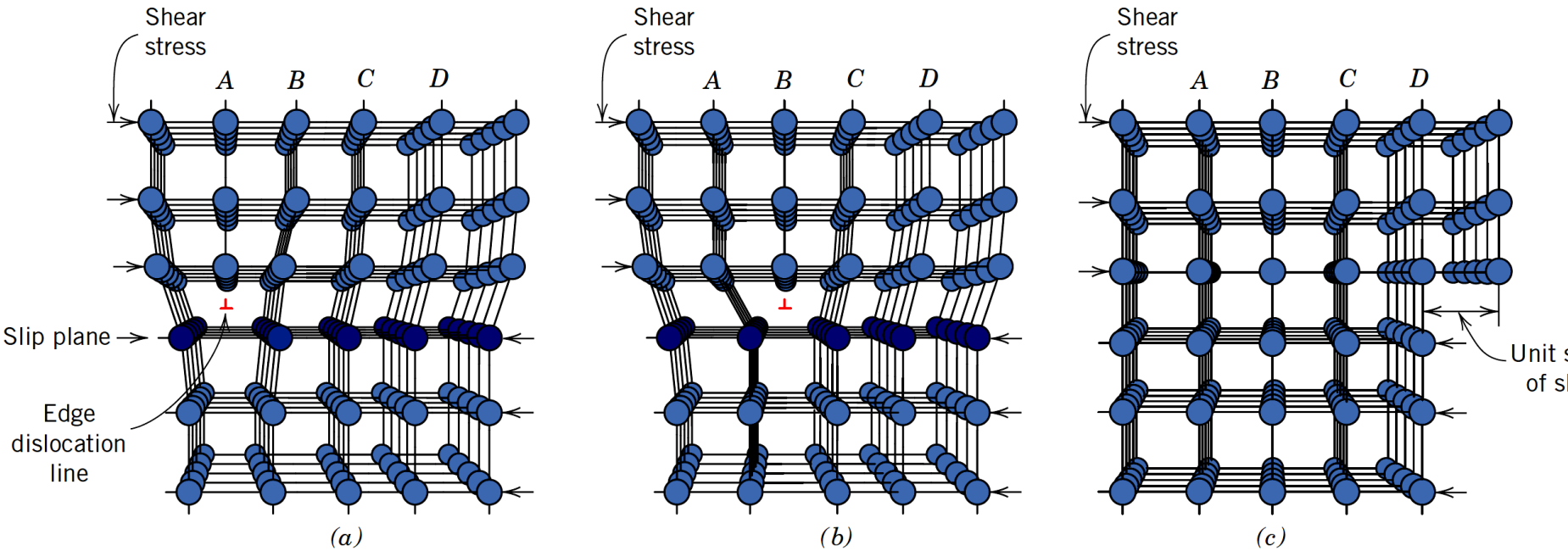


Figure: Motion of an edge dislocation.

Line defects

Dislocation movement

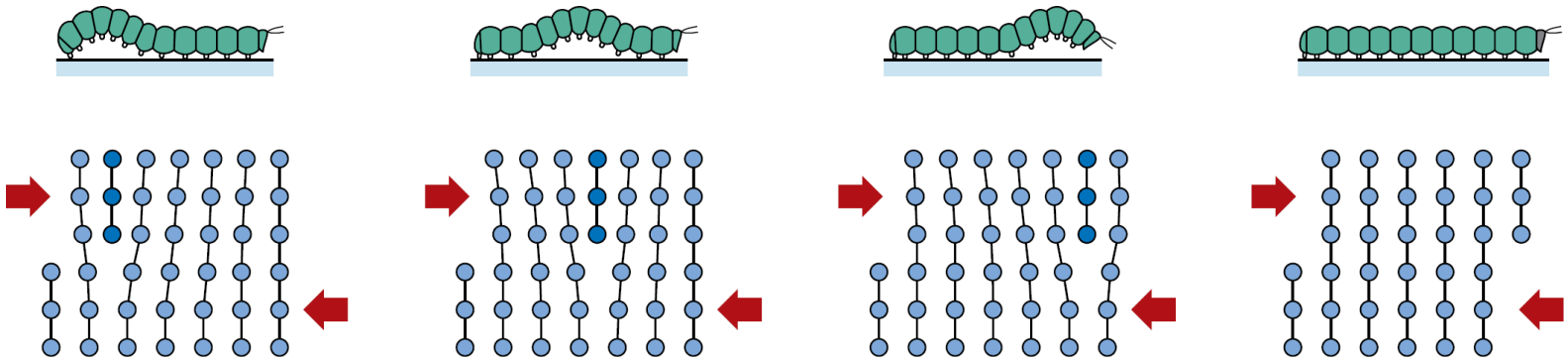
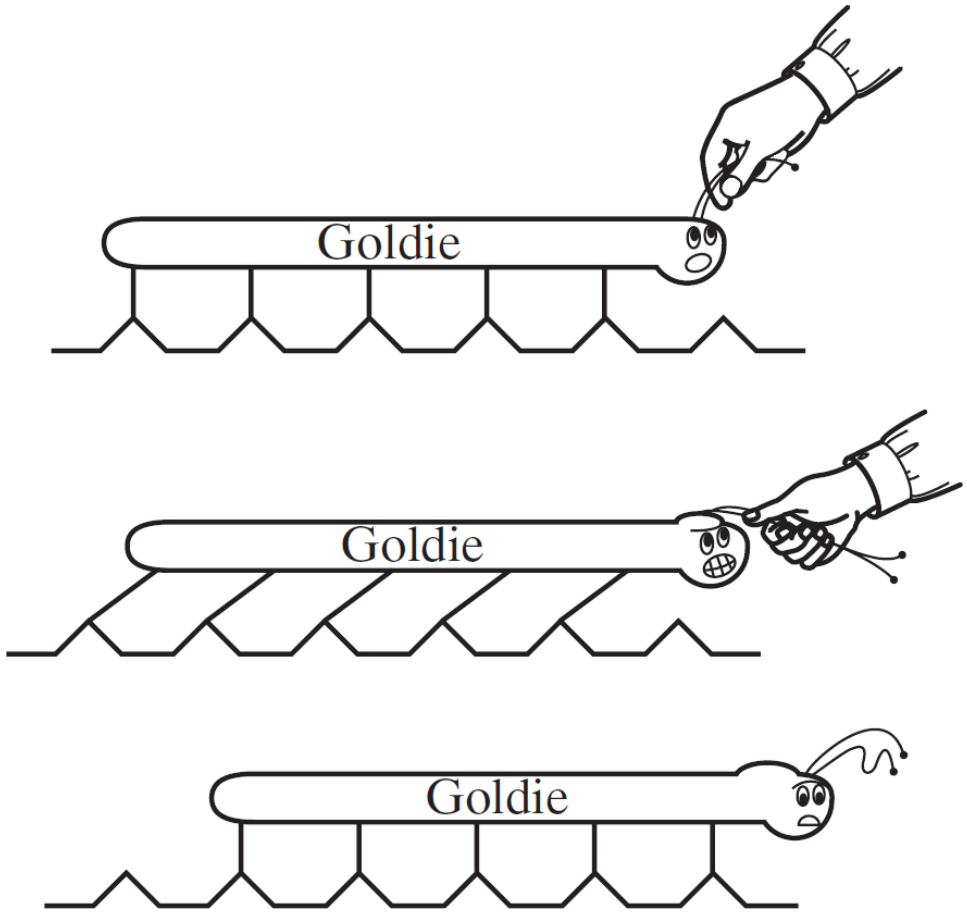


Figure: Movement compared to that of a catapillar

Line defects

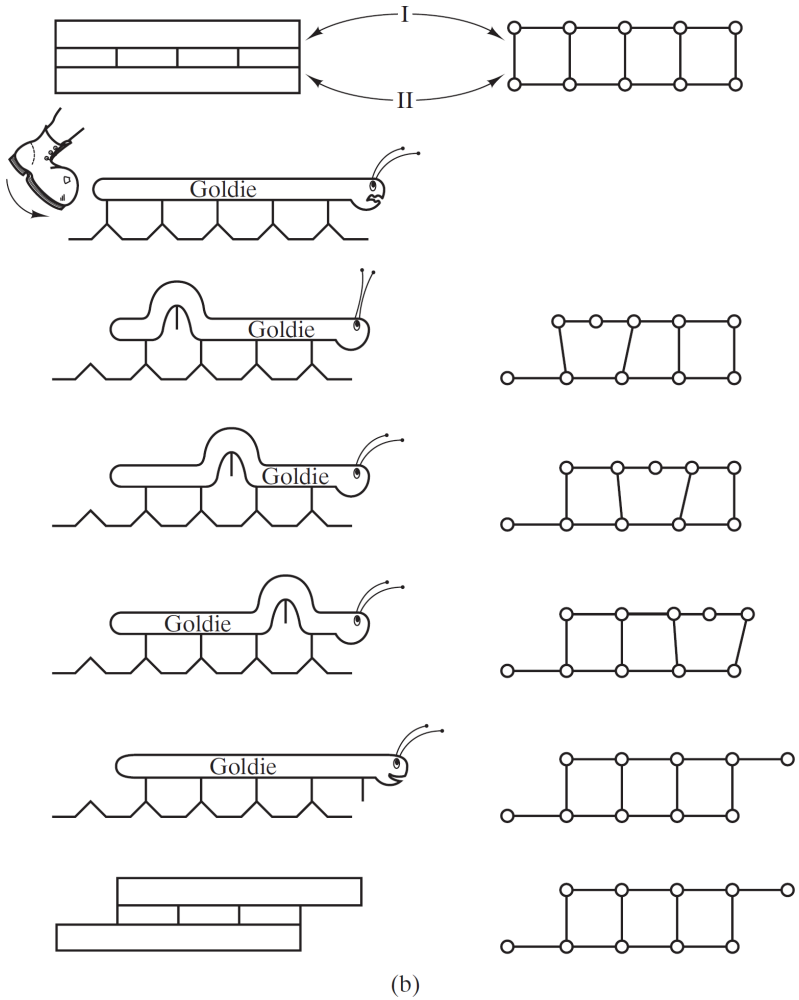
Dislocation movement



(a)

Line defects

Dislocation movement



(b)

A Burgers vector b is required to close a loop of equal atom spacings around the dislocation.

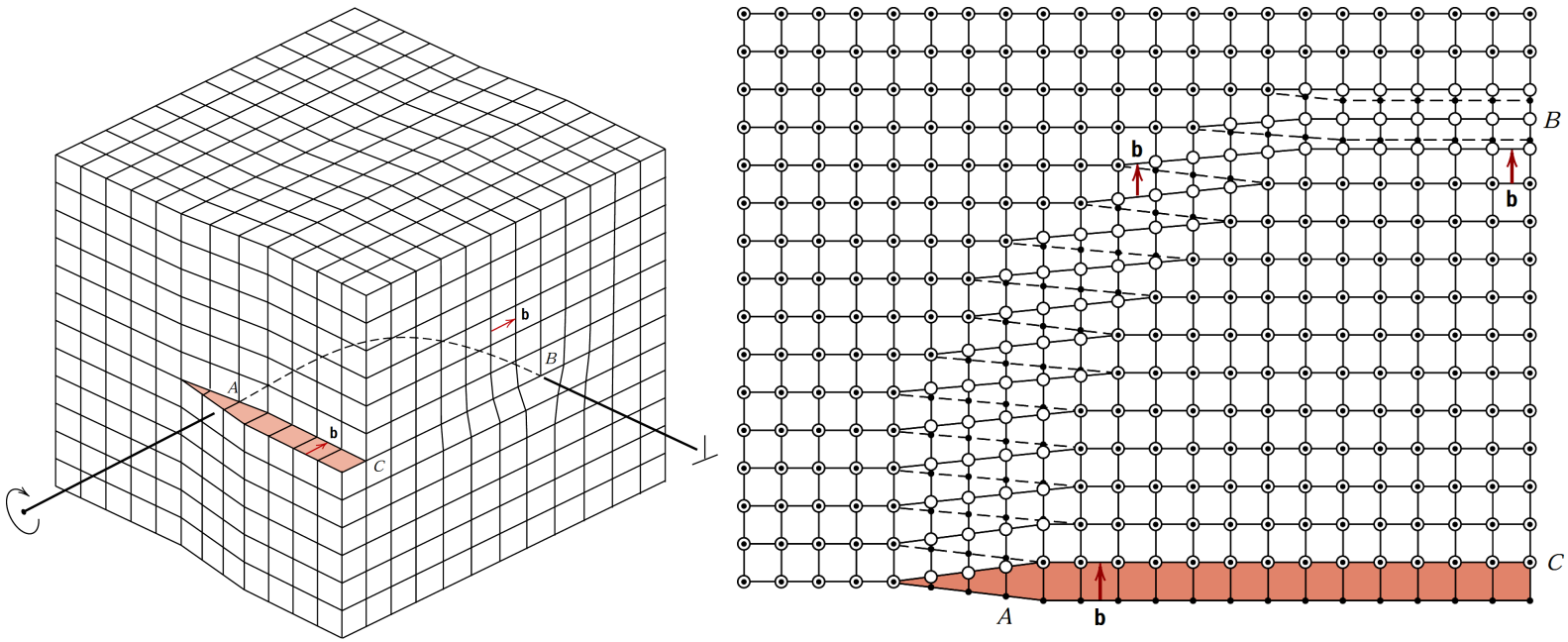
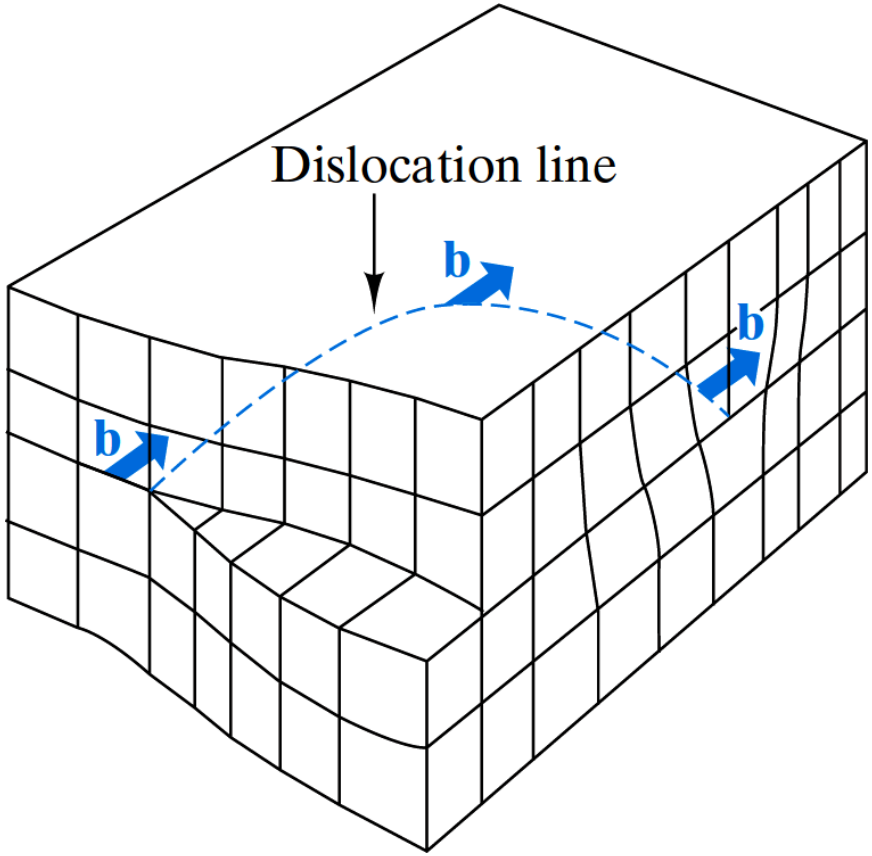
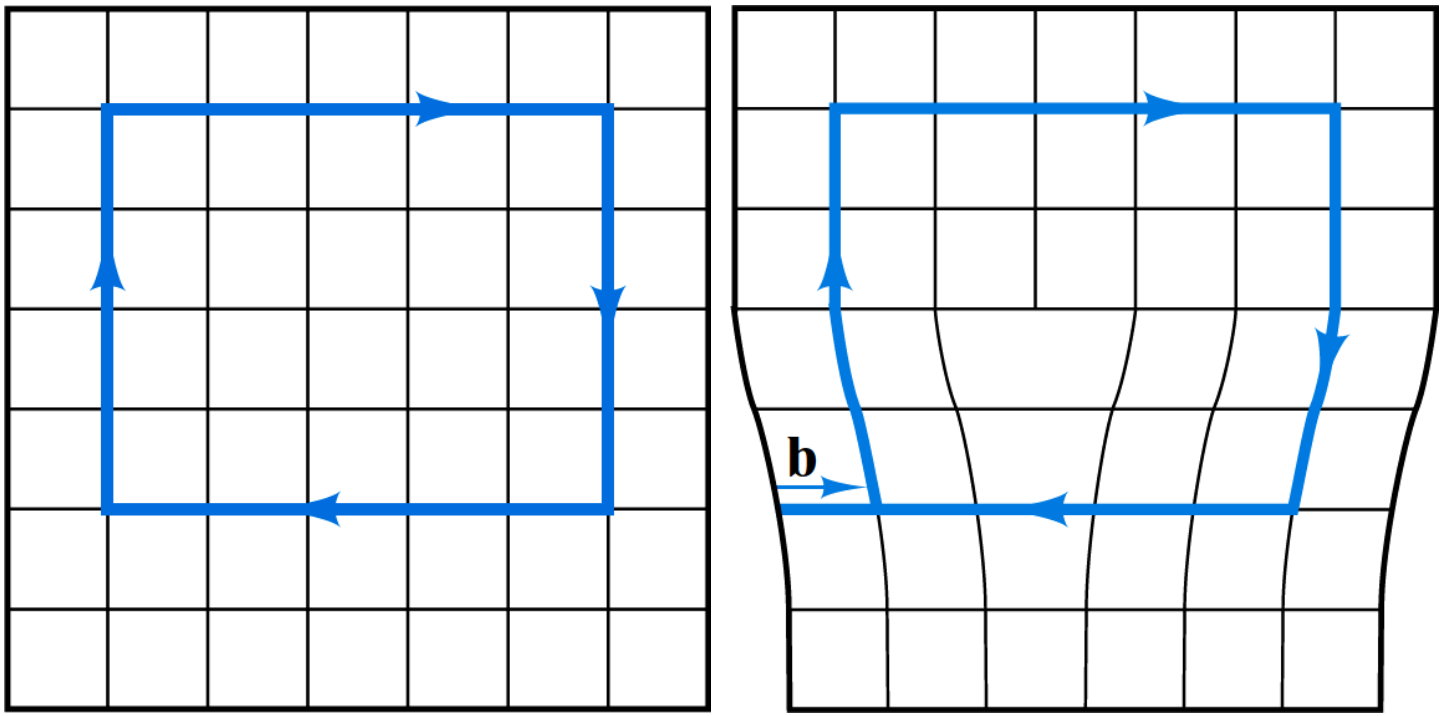


Figure:

A Burgers vector b is required to close a loop of equal atom spacings around the dislocation.

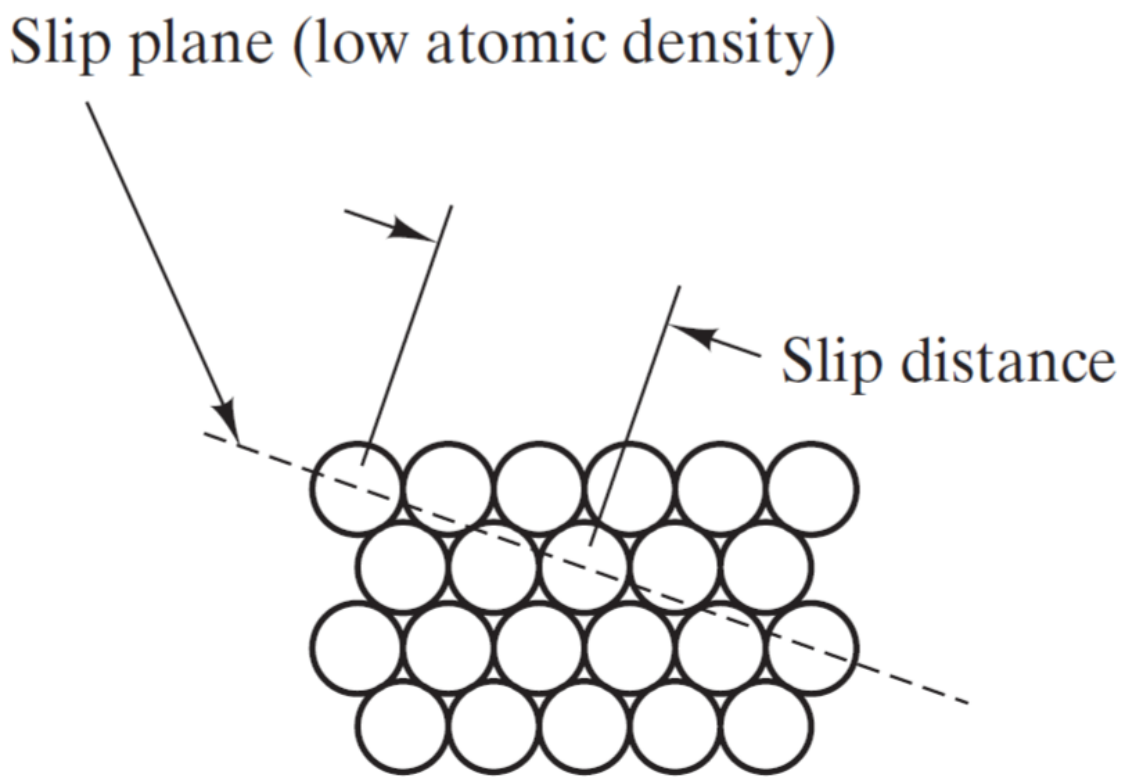


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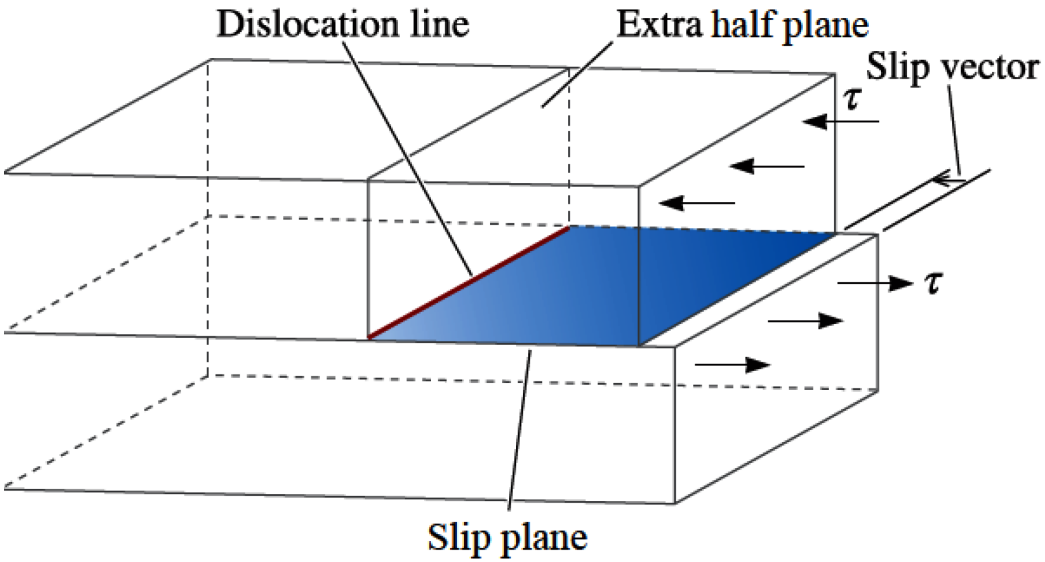
Dislocation motion

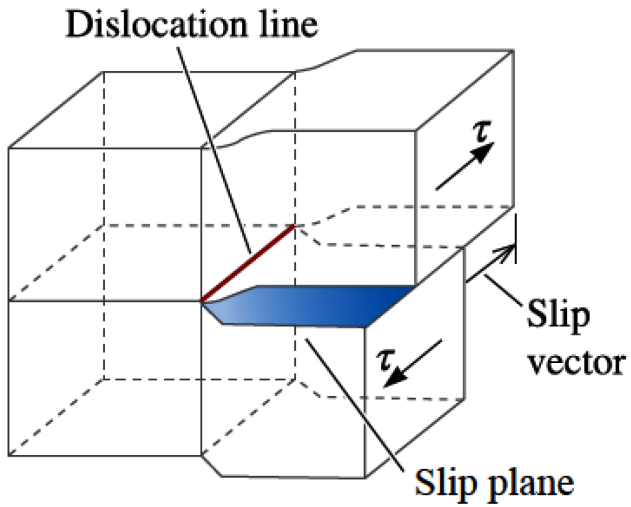
- Dislocation move through the crystal until it produces a step outside the crystal in the slip direction
- A slip system has a slip plane and a slip direction
- slip occurs only on high density planes and along directions of high linear density



Dislocation motion

In **edge** dislocations the dislocation line moves in a direction **parallel** to the **Burgers vector** while in **screw** dislocations it is **perpendicular**.

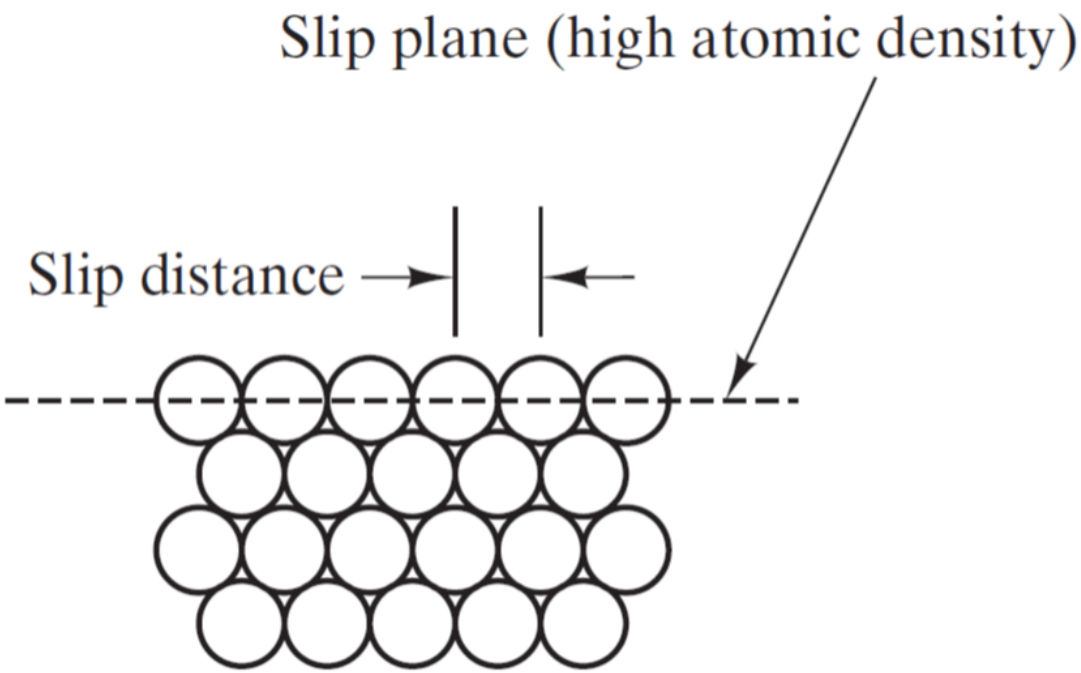




Dislocation motion

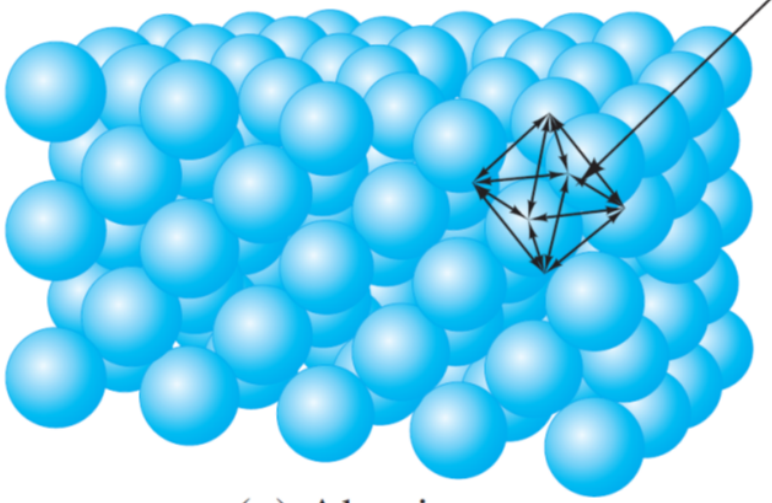
Significance of Dislocations

- **Plastic deformation** refers to irreversible deformation (change in shape) that occurs when the stress that caused it is removed.
- The applied **stress** moves dislocations.
- Permanent deformation occurs by the moved dislocations.



Dislocation motion

Slip system of Al



(a) Aluminum

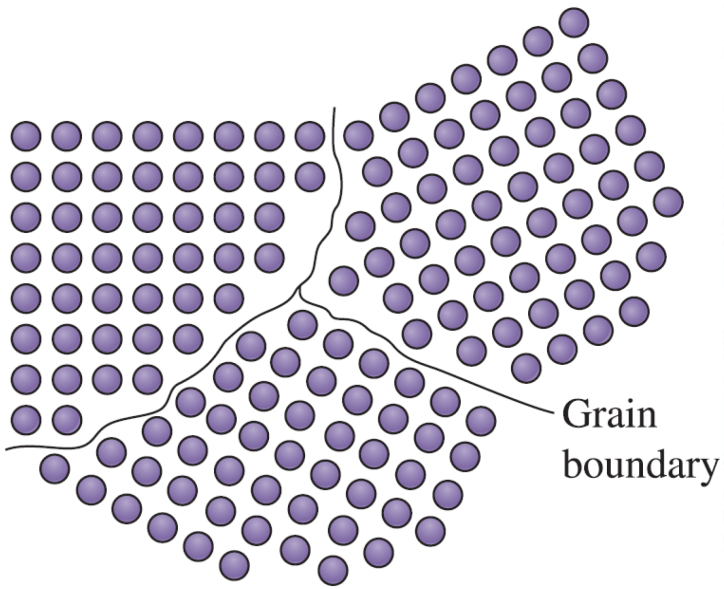
$\{111\}\langle 1\bar{1}0\rangle =$

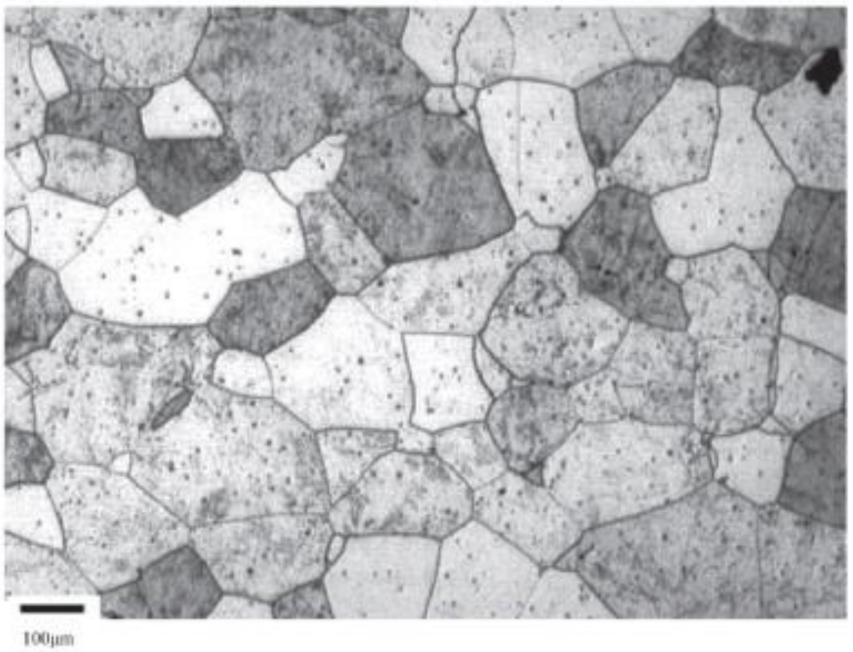
$(111)[1\bar{1}0]$	$(111)[\bar{1}01]$	$(111)[01\bar{1}]$
$(11\bar{1})[1\bar{1}0]$	$(11\bar{1})[101]$	$(11\bar{1})[011]$
$(1\bar{1}1)[110]$	$(1\bar{1}1)[10\bar{1}]$	$(1\bar{1}1)[011]$
$(\bar{1}11)[110]$	$(\bar{1}11)[101]$	$(\bar{1}11)[01\bar{1}]$

- The slip process, is important to understand the mechanical behavior of metals.
- Slip explains why the strength of metals is much lower than the value predicted from the metallic bond.
- Slip provides ductility (ability of materials to be permanently stretched or bent without breaking) in metals
- The mechanical properties of a metal or alloy can be controlled by making the movement of dislocations difficult.

Surface defects

- Surface defects are the boundaries, or planes, that separate a crystalline material into regions.
- For example a region of a material may have the same crystal structure but different orientations.





Surface defects

- The outside surface of the material is a place where the crystals (grains) suddenly ends.
- Atoms at the surface does not have the right coordination number and therefore atomic bonding is incomplete.
- The external surface may also be very rough, may have tiny notches, and may be much more reactive than the bulk of the material.

